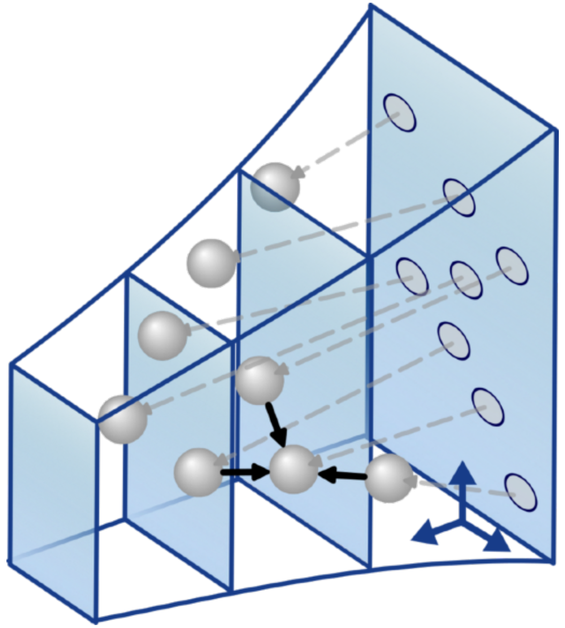
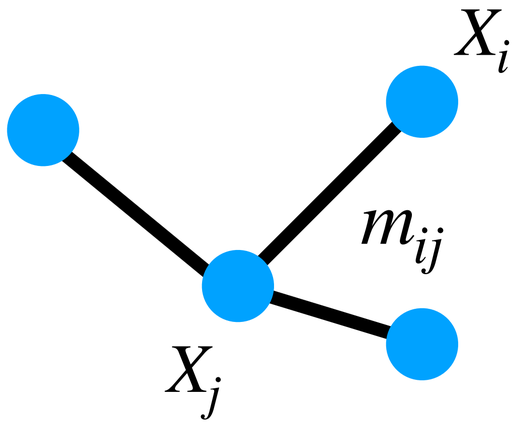
AdS-GNN: Bulk operations

Now we have a graph of points living in AdS space.



Act on it with **message-passing**; points talk to their neighbours in a way that depends only on intrinsic AdS proper distance.



Final information aggregated over AdS or sent back to boundary: **conformally equivariant neural network**.

AdS GNN: SuperPixel MNIST

- We test it on **SuperPixel MNIST** (Monti 2017), a dataset where each entry has a point cloud of 75 pixels and the network has to find the digit.

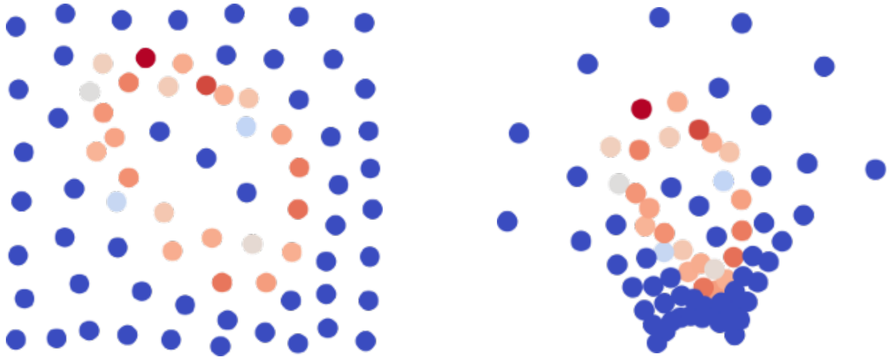


Table 1: Classification error on SuperPixel MNIST. We augment the test set with random rotations and scaling. ✗ indicates random-guess performance.

Model	Error rate, %		
	non-augmented	rotated	rotated+scaled
MONET	8.89	✗	✗
SplineCNN	4.78	✗	✗
GCGP	4.2	✗	✗
GAT	3.81	✗	✗
PNCNN	1.24 ± 0.12	✗	✗
PΘNITA	1.17 ± 0.11	1.17	✗
EGNN	4.17 ± 0.45	4.17	✗
AdS-GNN (**Ours**)	4.09 ± 0.27	4.09	4.09

Ours is comparable to existing ones, but exactly invariant under scaling.

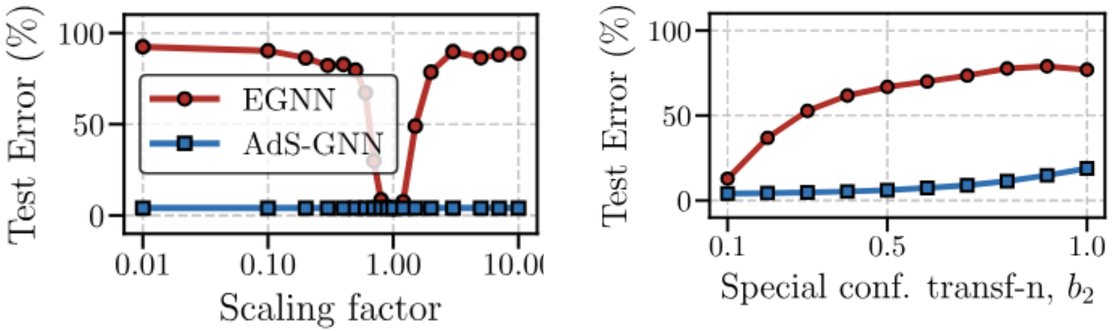



Figure 2: Test error on augmented data, SuperPixel MNIST.